ELIGULI HAN

Mathematics · Computation · Vertebrate Paleontology

New York University · Mathematics & Computer Science Undergraduate

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CV and project portfolio: https://eliguli712.github.io/Antiterrastrae/cv.html

Mathematics and computer science undergraduate developing computational approaches to animal anatomy, acoustics and environmental perception. Research interests include fossil cetacean cranial anatomy, temnospondyl biomechanics, and the connection between three-dimensional geometry and biological function.

Education & Mathematical Training

New York University · College of Arts & Science | Degree in progress

B.A. candidate, Mathematics & Computer Science

Completed coursework: Numerical Analysis; Partial Differential Equations; Machine Learning; Natural Language Processing. **Current coursework:** Mathematical Statistics.

Awesome SummerCamp: Three consecutive participations in advanced mathematical enrichment covering geometry, algebra and number theory.

Research Report

Han, E. (2026). Tooth-Level Finite-Element Analysis of Serration-Scale Stress Concentration in an Otodus megalodon Anterior Tooth. A µCT-to-Mesh Workflow with Bounded Functional Interpretation.

DURF Final Technical Report | Faculty advisor: Leslie Greengard

- Developed a µCT/DICOM-to-mesh workflow combining segmentation, topology repair, TetWild/Gmsh tetrahedral meshing, curvature checks and COMSOL finite-element analysis of tooth-scale stress localization under stated loading and material assumptions.

Read full technical report (PDF, 26 pages)

Selected Research & Computational Projects

Antiterrastrae · Volumetric Biosonar & Scientific Visualization | Ongoing

Self-directed computational prototype · JavaScript, HTML/CSS, WebGL

- Developing an interactive aquatic simulation connecting three-dimensional animal surface geometry with acoustic propagation, echoes, and temperature-, salinity- and depth-dependent environmental structure.
- Implemented a tetrahedral finite-element module for scalar acoustic eigenmodes, $Kp = \lambda Mp$, with stiffness and consistent mass matrices, Cholesky transformation and Jacobi eigenvalue iteration.
- Integrated WebGL echo rendering with a Canvas fallback, adjustable acoustic parameters, and VTK/OBJ export of surface and volumetric geometry for inspection and further analysis.

Cynthiacetus peruvianus · Digital Cranial Anatomy | Ongoing

Independent project in development · MNHN holotype MNHN.F.PRU10

- Initiated data-access correspondence with Guillaume Billet at the Muséum national d'Histoire naturelle; availability of cranial and mandibular surface scans was confirmed, subject to a formal collections request and use conditions.
- Developing a staged research plan linking surface morphology to CT/µCT segmentation, virtual endocast reconstruction and exploratory cranial biomechanics, according to available data and documented anatomical assumptions.

Coronodon Dental Morphology · Primary-Literature Verification | 2026

Independent research-verification assessment submission

- Constructed a source-verifiable question on fossil whale dental measurements; cross-checked interpretations against primary literature and documented the evidence behind a reference answer and model-error diagnosis.

Technical Skills & Research Direction

Programming: Python, NumPy, pandas, scikit-learn; JavaScript, HTML/CSS; WebGL/Canvas; SQL; GitHub; AI-assisted development and debugging.

Methods: Numerical methods, differential equations, finite-element assembly, eigenvalue problems, three-dimensional mesh processing, scientific visualization and literature verification.

Training sought: Specimen-based CT/µCT segmentation, comparative cranial anatomy, geometric morphometrics and finite-element analysis of biological structures.

Seeking supervised undergraduate research · Summer 2027 · New York · F-1 international student